

Giona Sala

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Research experience

PhD Rheinisch-Westfälische Technische Hochschule (RWTH) Aachen, Aachen, DE
 Doctorate in Theoretical Physics Oct 2023 – present
 • Topic: *Gravitational Wave Cosmology*
 • Supervisor: Julien Lesgourgues

Education

MSc École Polytechnique Fédérale de Lausanne (EPFL), Lausanne, CH
 Master of Science in Physics Sep 2020 – Apr 2023
 • Focus on Theoretical Particle Physics
 • Minor in space technologies
Université de Genève (UniGe), Geneva, CH
 Master's thesis Sep 2022 – Apr 2023
 • Title: *Implications of Primordial Black Holes physics at high red-shifts*
 • Supervisor: Antonio Riotto
BSc Eidgenössische Technische Hochschule Zürich (ETHZ), Zürich, CH
 Bachelor of Science in Physics Sep 2017 – Sep 2020
 • Final project on Hardy's Paradox
University of Toronto (UofT), Toronto, CA
 Exchange Sep 2019 – Dec 2019

Publications

Detecting White Dwarf Binary Mergers with Gravitational Waves Jun 2026
Giona Sala, Chiara Brandenstein, Sebastian Baum, Peter W. Graham
[Phys. Rev. D 113 \(2026\) 123022](#) [↗](#)
Inferring cosmological parameters from galaxy and dark sirens cross-correlation May 2026
Giona Sala, Alessandro Cuoco, Julien Lesgourgues, Kostantinos-Rafail Revis,
 Lorenzo Valbusa Dall'Armi, Santiago Casas
[JCAP 05 \(2026\) 095](#) [↗](#)

Conference contributions & talks

Swiss Cosmology Days 2026 Geneva, CH
 Jun 2026
 • Presentation on *Pushing cosmological inference with next-generation gravitational wave surveys*
3rd Sardinian Gravitational Wave Science (SGWS) International Mini-Workshop Cagliari, IT
 May 2026
 • Presentation on *Can we detect the anisotropies of the stochastic gravitational waves background?*
Gran Sasso Science Institute (GSSI) seminar L'Aquila, IT
 Feb 2025
 • Invited talk on *The Hubble tension in the GW era: A Cross-Correlation study*
Seminar on the Interface of Particle Physics and Gravitational Waves Online
 Jan 2025
 • Presentation on *Detecting White Dwarf Binary Mergers with Gravitational Waves*

MAGIS-100 meetings

- Presentation on *Detecting White Dwarf Binary Mergers with Gravitational Waves*

Online (Fermilab, US)
Nov 2025

Invisibles25 Workshop

- Presentation and poster on *Inferring cosmological parameters from galaxy and dark sirens cross-correlation*

CERN, CH
Sep 2025

WE-Heraeus-Seminar: new windows on the Universe

- Poster on *Inferring cosmological parameters from galaxy and dark sirens cross-correlation*

Kitzbühel, A
May 2025

European Consortium for Astroparticle Theory (EuCAPT) Annual Symposium

- Plenary lightning talk on *Inferring cosmological parameters from galaxy and dark sirens cross-correlation*

Online (CERN, CH)
May 2025

Additional relevant experiences

Teaching experience

RWTH,
Teaching assistant

Aachen, DE
Apr 2024 – present

- Supervision:
 - Master's thesis on *Component separation algorithms for the gravitational wave backgrounds*
 - Bachelor's thesis on *Separation of the cosmological and astrophysical gravitational wave background*
- Courses: Theory of Relativity and Cosmology x2; The ingredients of the universe; Theoretical Physics I: Mechanics

EPFL,
Teaching assistant

Lausanne, CH
Sep 2021 – Jan 2023

- Courses: General Physics: Electromagnetism x2, General Physics II, Experimental Physics I, Physics IV

Work experience

Fondazione Istituto Ricerche Solari (IRSOL),
Civil Service

Locarno-Monti, CH
Apr 2023 – Aug 2023

- Solar research technical and scientific collaborator

Scuola Universitaria Professionale della Svizzera Italiana (SUPSI),
Civil Service

Mendrisio, CH
Jun 2022 – Jul 2022

- Renewable energy research collaborator

Extracurricular activities

EPFL Spacecraft Team (EST),
System engineer team leader

Lausanne, CH
Sep 2021 – Jun 2022

- Student association that coordinates the CHESS (Constellation of High Energy Swiss Satellites) space mission

EPFL Rocket Team (ERT),
Payload team leader

Lausanne, CH
Sep 2020 – Oct 2021

- Team of students from EPFL building Bella Lui II 2021 rocket for international competitions

Akademische Raumfahrt Initiative Schweiz (ARIS),
Structure engineer support

Zürich, CH
Feb 2020 – Jul 2020

- The student association for space in the German part of Switzerland, building EULER 2020 rocket

Skills

Programming languages

Python	●●●●●	C/C++	●●●●●
LaTeX	●●●●●	MATLAB	●●●●●

Programmes

Mathematica	●●●●●	MontePython	●●●●●
Excel	●●●●●	CLASS	●●●●●

Languages

Italian	Native	French	●●●●●
English	●●●●●	German	●●●●●

References

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Institute for Theoretical Particle Physics
and Cosmology
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Department of theoretical physics
Université de Genève
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Peter W. Graham

Prof. Dr.
Leinweber Institute for Theoretical Physics
Stanford University
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Sebastian Baum

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Mid Sweden University
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